
Geo-Octo: Infusing VLAs with Spatial Information to Improve 3D-Task Performance

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Abstract

Recent advances in robot learning have centered on Vision-Language-Action (VLA) models as a reliable method for training generalist instruction-following agents. However, despite their significant potential, most VLAs rely on backbones pre-trained fundamentally on 2D data. This presents a critical limitation: while robotic actions are inherently executed in 3D space, the model’s understanding is limited in two dimensions, constraining its spatial reasoning capabilities. To bridge this gap, we leverage recent advancements in monocular 3D foundation models—specifically the Visual Geometry Grounded Transformer (VGGT)—to guide the action prediction process. We introduce Geo-Octo, a framework designed to infuse Octo, a VLA known for its flexible architecture, with geometric knowledge from VGGT. To determine the most effective method for incorporating 3D information, we experiment with architecture shifts based on latent feature fusion as well as readout-level injection. We evaluate these methods on LIBERO benchmark suites and conclude that the injection of 3D data in VLAs is not trivial. The results suggest that 2D-pretrained models are surprisingly resistant to incorporating 3D geometry via simple finetuning due to a strong, inherent 2D bias. The code is publicly available at: <https://github.com/Platon-7/Geo-Octo/>.

1 Introduction

Vision-Language-Action (VLA) models represent a shift toward unified, generalist robot learning. VLAs are end-to-end systems that translate visual and linguistic inputs into low-level control actions. This approach adapts the broad semantic knowledge of Vision-Language Models (VLMs) for robot manipulation, using specialized architectures to bridge visual reasoning and actions. By leveraging large-scale data, these models have demonstrated remarkable success in handling diverse manipulation tasks, establishing them as a robust solution for robotics (Black et al., 2024; Kim et al., 2024).

However, a limitation of these models lies in their pretraining representations. Most VLA backbones are initialized from VLMs trained on large-scale 2D image datasets. Consequently, their internal features are rooted in 2D pixel space. This becomes a bottleneck in robotics, where manipulation is an inherently geometric task requiring precise interaction in 3D space. The model recognizes objects semantically, but it often lacks the intrinsic spatial awareness required for fine-grained control.

An intuitive solution would be to pretrain foundation models directly on large-scale 3D datasets, such as point clouds or voxel grids. High-quality 3D data is significantly scarcer and more expensive to collect than 2D images though, making this approach impractical for many researchers. Instead of discarding the capabilities of current 2D models, a more resource-efficient alternative is to equip existing agents with geometric knowledge. We propose to infuse a generalist VLA with 3D spatial

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understanding distilled from a pretrained expert vision model, thus bridging the modality gap without the prohibitive cost of pretraining from scratch.

We introduce Geo-Octo, a framework designed to investigate this knowledge transfer. We select Octo (Ghosh et al., 2024) as our base policy because of its flexible architecture, which allows different multimodal inputs to be processed as a unified sequence. To source the geometric data, we employ the Visual Geometry Grounded Transformer (VGGT) (Wang et al., 2025a), a model capable of predicting dense 3D structures from single RGB images. However, determining which aspect of this geometric information—implicit latent features or explicit point maps—is most compatible with a VLA is non-trivial. Therefore, we explore distinct integration strategies: fusing compressed intermediate features at the tokenizer level, relying solely on VGGT’s geometric representations as the exclusive visual input, and injecting 3D pointmaps directly into the action readout heads.

In summary, our contributions are as follows:

- We experiment with various injection points for 3D data within the VLA architecture, evaluating methods based on latent feature fusion, exclusive reliance on geometric representations, and readout-level gating.
- We perform an in-depth analysis of the compatibility between 2D-pretrained backbones and 3D geometric data. We find that VLAs possess a strong inductive bias towards 2D patterns, making them resistant to incorporating geometric information through simple finetuning.

2 Related Work

Generalist Policies & Vision-Language-Action Models. Recent progress in robot learning is driven by a shift from specialized agents to generalist policies trained on large-scale datasets (Ghosh et al., 2024; Collaboration et al., 2023; Brohan et al., 2023; Wang et al., 2024b). Adapting web-scale VLMs into VLA models has facilitated this trend. Other works (Zitkovich et al., 2023) established that pretrained VLMs can condition on language and vision to generate actions directly. Scaling this approach has produced a new generation of capable agents, including OpenVLA (Kim et al., 2024) and PI0 (Black et al., 2024). Optimization improvements have led to refined follow-ups like PI0.5 Intelligence et al. (2025) and OpenVLA-OFT (Kim et al., 2025), alongside recent state-of-the-art models like RDT-1B (Liu et al., 2025) and GR00T (Bjorck et al., 2025). Yet, despite their impressive generalization, these architectures process inputs primarily as sequences of 2D image patches. This reliance on 2D representations motivates our work. We investigate whether explicitly injecting 3D geometric information can enhance spatial reasoning for complex manipulation tasks.

Enhancing VLAs with 3D Geometric Priors. To address the spatial limitations of 2D VLAs, recent work seeks to infuse models with 3D awareness. Some approaches, such as PointVLA (Li et al., 2025a) and TraceVLA (Zheng et al., 2025), focus on lightweight adaptation, using modular injections or targeted finetuning to preserve pre-trained representations. Conversely, methods like SpatialVLA (Qu et al., 2025) and BridgeVLA (Li et al., 2025b), along with diffusion-based policies such as SEM (Lin et al., 2025), DexVLA (Wen et al., 2025), and FP3 (Yang et al., 2025), propose more fundamental shifts. These range from novel positional encodings to full-scale pretraining of 3D-native policies. While these efforts validate the importance of geometric cues, they typically rely on explicit 3D sensor inputs or computationally expensive pretraining. In contrast, we introduce Geo-Octo to investigate a complementary, resource-efficient direction: infusing 3D knowledge from a single-view geometry expert and transferring it to a VLA via lightweight token-level finetuning.

Learning 3D Scene Geometry from 2D Images. We source geometric knowledge from 3D computer vision, a field that focuses on inferring scene structure from 2D images. Specifically, we employ VGGT (Wang et al., 2025a). This state-of-the-art model predicts dense geometry zero-shot from a single uncalibrated image. It builds on extensive research in Structure-from-Motion (SfM) (Wang et al., 2024a; Zhang et al., 2025; Chen et al., 2025; Duisterhof et al., 2025; He et al., 2023), including concurrent works like (Elflein et al., 2025). Architecturally, it relates to recent Multi-View Stereo (MVS) models (Wang et al., 2024c; Leroy et al., 2024; Wang et al., 2025b; Tang et al., 2025; Cabon et al., 2025). A parallel paradigm captures structure implicitly via dense point tracking (Doersch et al., 2023; Karaev et al., 2024b,a). The success of these methods proves that high-fidelity 3D information can be robustly extracted from 2D images. This supports our central hypothesis: this geometric knowledge can be transferred to enhance the spatial reasoning of 2D-pretrained VLAs.

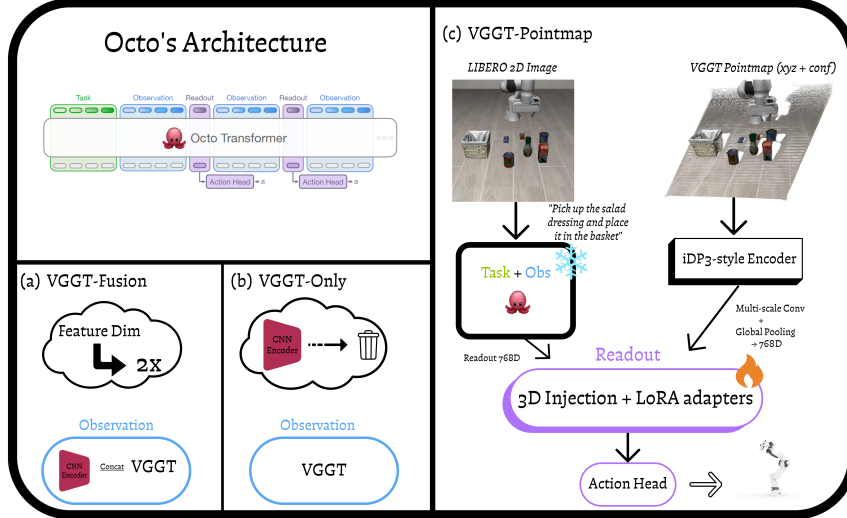


Figure 1: Overview of Octo-based finetuning variants. (a) VGGT-Fusion doubles the observation feature dimension by concatenating VGGT tokens with Octo’s patch embeddings. (b) VGGT-Only discards the vision encoder and feeds only VGGT tokens into the transformer. (c) VGGT-Pointmap keeps the Octo transformer frozen, encodes VGGT pointmaps with an iDP3-style CNN, and injects the resulting 768-D embedding into the readout via trainable LoRA adapters before the action head.

3 Methodology

This section details the methodology for Geo-Octo, our framework for investigating the infusion of 3D geometric knowledge from a pretrained vision expert into a 2D-pretrained VLA model. Our central approach involves using the frozen, pretrained VGGT to generate 3D-aware tokens, which are then used to finetune the Octo model. We explore three distinct strategies for this knowledge transfer, distinguished by the source of the VGGT tokens as well as their injection point: one using features from VGGT’s intermediate layers, one that replaces completely the pretrained vision encoder, and finally, another using its final, explicit pointmap output. Before delving into the various methods, we will first describe the main pipeline in Section 3.1.

3.1 The Octo Architecture

Octo’s policy stacks its inputs in three stages: (i) modality-specific tokenizers that turn raw observations—RGB frames, proprioception, language—into fixed-length token grids, (ii) a projection layer that lifts those tokens into the common transformer width of 768, and (iii) a frozen decoder-only transformer that produces action readouts. For all our experiments, we leverage the Octo-Base model. For Octo-Base, there is a tensor of shape $(B, 1, 256, 512)$, where B represents the batch size, 1 corresponds to the temporal window, 256 is the number of spatial patch tokens, and each of those tokens has 512 tokens. After the observation tokenizer, a learned linear map expands those 512 channels to match the transformer width, 768, before the sequence is handed to the backbone.

Across all our proposed methods, components outside the primary vision and action-readout pathways remain identical to Octo’s released model. Language instructions, task-conditioning tokens, and proprioceptive features are inherited without modification. Our approaches diverge primarily in where and how the geometric information is integrated. The first two strategies—token fusion and replacement—intervene at the observation tokenizer stage, preparing VGGT-derived tokens to be processed by the existing projection layer. In contrast, our third strategy, the pointmap variant, takes a different approach. It introduces 3D information much later in the pipeline, fusing it directly with the action readout tokens after they have been processed by the transformer backbone. This variant is also distinct in its training recipe, utilizing Low Rank Adaptation (LoRA) (Hu et al., 2021) on a mostly frozen model. The subsequent sections will detail these distinct preparation and integration strategies, outlining the specific architectural choices made for each.

3.2 VGGT-Fusion Using Intermediate Layers

Our first approach is based on the hypothesis that for imparting a general 3D understanding to a robotic policy, the latent features from VGGT’s intermediate layers (prior to these task-specific heads) may be more valuable. So, instead of using any of the direct 3D outputs that VGGT provides, such as specialized prediction heads, depth maps, point maps, or keypoint tracking, we aim to leverage these rich intermediate implicit geometric features to enhance Octo’s spatial reasoning capabilities.

To achieve this, we extract features from all 24 intermediate layers of VGGT and fuse them into Octo. This requires a specific pipeline to transform the VGGT features into a format compatible with Octo’s vision encoder. The integration process proceeds in four sequential stages: (1) image preprocessing to align inputs with VGGT’s patch grid; (2) feature extraction to retrieve latent states from intermediate layers; (3) feature compression via a learned autoencoder to reduce dimensionality; and (4) token fusion, where these geometric embeddings are concatenated with Octo’s native visual tokens.

Image Preprocessing for VGGT. To ensure compatibility with the VGGT expert, each input image is transformed into a 224×224 tensor. Following standard RGB conversion, the images undergo an aspect-preserving resize, ensuring dimensions align with the model’s patch grid (multiples of 14). White padding is applied to achieve the target square resolution. Finally, the pixel values are normalized to the $[0, 1]$ range and the tensor is permuted to the channel-first ($C \times H \times W$) format.

Feature Extraction and Compression via an Autoencoder. Subsequently, the preprocessed image is fed into the frozen VGGT model. We extract the feature map from each one of its 24 layers, resulting in a raw feature tensor of shape $(24, 256, 2048)$, where 24 is the number of layers, 256 is the number of spatial tokens $((224/14) \times (224/14))$, and 2048 is the feature dimension. This tensor is processed by an autoencoder that we pretrain for this compression task. The encoder stage first applies a weighted mean pooling across the 24 layers to produce a $(256, 2048)$ tensor. This is followed by an MLP layer that maps the resulting vector to the target bottleneck shape of $(256, 512)$.

Autoencoder Architecture and Training. Architecturally, the module begins with a learnable layer-wise aggregation mechanism designed to pool the 24 intermediate feature maps from VGGT into a unified representation of shape $(256, 2048)$. Both the encoder and decoder utilize GELU activation functions and apply LayerNorm to their input and bottleneck layers. The MLP has a hidden width of 8192, and the bottleneck is designed to compress each of the 256 spatial tokens into a target 512-dimensional feature vector. To train this autoencoder, we constructed a dataset comprising images from all four splits of the LIBERO benchmark (Object, Spatial, Goal, o10) (Liu et al., 2023). To ensure its robustness for the downstream finetuning task, we applied the same workspace-style augmentations used in Octo’s original pretraining, including random resized crops, brightness, contrast, saturation, and hue adjustments. The model was trained for 25 epochs using the Adam optimizer with a learning rate of 1×10^{-4} , with the objective of minimizing the Mean Squared Error (MSE) between the decoder’s output and the fused per-token feature target from VGGT.

Final Token Fusion. The output of the autoencoder is a feature tensor of shape $(256, 512)$. During the finetuning of the main policy, this tensor is concatenated along the feature dimension with Octo’s native vision tokens (which are also of shape $(256, 512)$). This operation transforms Octo’s vision input from $(B, 1, 256, 512)$ to a new, fused tensor of shape $(B, 1, 256, 1024)$. This effectively doubles the feature information available to the model’s attention layers for each spatial token. Note that, as we mentioned in Section 3.1, this feature dimension will be projected to the specified transformer input size, which is 768, so ultimately the transformer’s input will be the same dimension-wise.

3.3 VGGT-only

Our second approach tests a stronger and more architecturally distinct hypothesis. We aim to investigate whether the geometric features extracted from the VGGT tokens are comprehensive enough to entirely replace Octo’s native 2D vision encoder. The objective of this method is to determine if a policy grounded exclusively in 3D geometric information, without direct access to the original RGB pixel data, can successfully perform complex manipulation tasks.

The implementation pipeline is nearly identical to our first approach (Section 3.2), utilizing the same image preprocessing and pretrained autoencoder to generate a $(256, 512)$ feature tensor from VGGT’s intermediate layers. However, the key difference lies at the point of fusion. Instead of concatenating this feature tensor with Octo’s vision tokens, we discard Octo’s standard vision encoder entirely.

The feature tensor from our autoencoder becomes the sole visual input to the transformer’s attention layers. This configuration is more lightweight during finetuning, as it bypasses Octo’s native image patch encoder. The vision channel that the transformer processes is derived purely from VGGT’s latent geometry, creating a policy that is independent of color during its decision-making process and relies exclusively on the distilled 3D representation.

3.4 Pointmap Token Fusion

In contrast to using implicit latent features, this third strategy investigates the infusion of explicit 3D structural measurements from VGGT’s output heads. The hypothesis is that providing the Octo model with direct, metric 3D point map data will offer clearer and more unambiguous spatial cues for manipulation tasks. The objective is to convert VGGT’s per-pixel world-point predictions into a compact embedding and inject it into Octo’s action generation pathway without disrupting the primary visual processing pipeline.

Image Preprocessing and Feature Extraction. The image preprocessing pipeline is identical to the one described in Section 3.2, transforming input frames into a 224×224 tensor. These frames are passed through the frozen VGGT model, and we extract the output from its *world points* prediction head. We concatenate VGGT’s per-pixel confidence prediction to this output, resulting in a dense pointmap tensor of shape $(B, 1, 224, 224, 4)$, where B corresponds to the batch size and 1 to the timestep dimension. The final dimension represents the (x, y, z) coordinates and a confidence score for each point. For robustness, we apply a normalization step, which subtracts the per-image mean from the xyz channels and divides by the standard deviation, leaving the confidence channel untouched.

Pointmap Encoder Architecture. To transform the dense pointmap into a compact representation suitable for fusion, we introduce a `PointMapEncoder` module within the Octo model. This iDP3-style (Ze et al., 2025) convolutional encoder, initially downsamples the pointmap tensor for each timestep by a factor of two using average pooling, reducing its spatial resolution and computational cost. The downsampled tensor is then processed by a series of three convolutional blocks, which include GELU activations and max pooling operations, to generate a multi-scale feature pyramid. The features from each scale of this pyramid are globally average-pooled to create a set of fixed-size feature vectors. These vectors are subsequently concatenated, passed through a `LayerNorm`, and finally projected via a dense layer to match the transformer’s embedding size (512 for Octo-Base). This encoder effectively distills the entire point cloud and its confidence distribution for each timestep into a single global embedding vector of shape $(B, 1, 512)$.

Gated Fusion with Readout Tokens. Instead of fusing with the visual patch tokens, the geometric embedding is integrated directly with Octo’s action readout tokens, allowing the 3D information to directly influence the action generation process. This is illustrated in Figure 1. The fusion mechanism is implemented as a lightweight, gated residual update. For each readout token, both the token and the global pointmap embedding are first passed through separate linear layers into a bottleneck dimension of 256. The outputs are then aligned using `LayerNorm` and fused additively. This fusion is controlled by a learnable, per-readout-token scalar gate, which allows the model to learn how much 3D information to incorporate for each action dimension. Finally, the resulting fused vector is projected back to the original 512-dimensional embedding space to complete the update.

During finetuning, the main transformer blocks of Octo remain frozen. Only the parameters of the new `PointMapEncoder`, the readout bottleneck adapters, the LoRA matrices, and the final output heads are updated. This preserves the powerful pretrained knowledge of the Octo backbone while efficiently adapting it to leverage the new 3D geometric input. The final architecture thus supplies the transformer with its original visual patch stream, while the action-generation process is enriched by a globally pooled, gated 3D point representation.

Our integration strategy is influenced by the approach in PointVLA (Li et al., 2025a), but is adapted to better suit our specific framework and hypothesis. Our method diverges from PointVLA in two architectural decisions. First, while their method operates on pointclouds, our system leverages the dense, per-pixel pointmap generated by VGGT. Second, and more critically, we alter the point of fusion. Instead of performing a skip block analysis to inject geometric information into the intermediate layers of the transformer, we fuse the information at the final readout stage of Octo. This simplifies the architecture and directly informs the action-generation process with 3D cues.

4 Experiments

In this section, we present the evaluation of our Geo-Octo framework. We first detail the experimental setup, including the benchmarks, baselines, and implementation specifics. We then present the main quantitative results. Finally, we provide a detailed analysis and a series of qualitative visualizations to diagnose the reasons for the performance gaps.

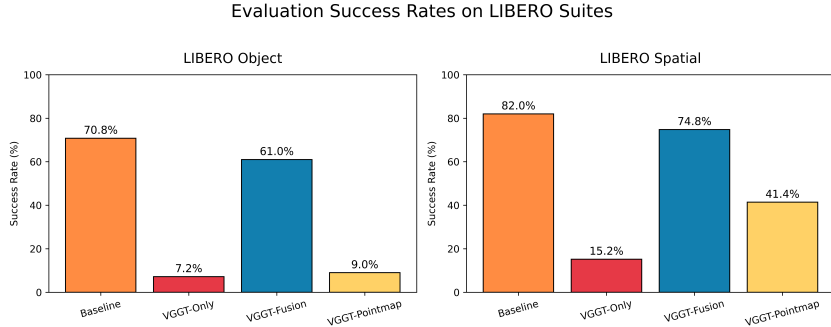


Figure 2: **Evaluation Success Rates on LIBERO Suites.** Comparing the proposed variants, VGGT-Fusion demonstrates the highest resilience, whereas VGGT-Only suffers catastrophic collapse. The pointmap variant shows notable sensitivity, performing significantly better on Spatial tasks. Results are reported from the best-performing checkpoint for each model.

4.1 Experimental Setup

Tasks and Evaluation. We evaluate all models on the LIBERO benchmark, a suite of long-horizon language-conditioned manipulation tasks. Our experiments focus on two splits: LIBERO-Object and LIBERO-Spatial. For each split, we evaluate performance across 10 distinct tasks. Each task is evaluated for 50 episodes, totaling 500 evaluation episodes per split for each method.

Baseline. Our baseline is the original Octo-Base model, finetuned on the LIBERO tasks using the official codebase and hyperparameters. This provides a direct point of comparison to assess the impact of our injected geometric tokens. We highlight that compared to Octo’s official code, we use a batch size equal to 64 instead of 256 due to GPU limitations. In order to ensure that the comparison is fair, we set gradient accumulation equal to 4 and multiply accordingly the total number of iterations (200,000 iterations in total). The employed learning rate was 3×10^{-4} .

Implementation Details. For the VGGT-Fusion and VGGT-Only methods, we used an identical script to the baseline. For the VGGT-Pointmap variant, which introduces a new pointmap encoder and uses LoRA, we adapted the training configuration following insights from PointVLA. Due to the smaller number of trainable parameters, we used a lower learning rate of 1×10^{-4} and a shorter training of 50k steps. We enabled LoRA for both attention (Q and V projections) and MLP layers within the transformer, with a rank of $r=8$, a scaling factor of $\alpha=16$, and no dropout. All core transformer weights remained frozen, so gradients were only applied to the LoRA matrices and the new pointmap processing layers. Due to computational constraints, we did not perform a hyperparameter search for any of the methods.

4.2 Results

Next, we evaluate how distinct integration strategies affect the VLA’s ability to leverage geometric cues. In Figure 2 we observe a summary of the performance of the three proposed Geo-Octo variants.

VGGT-Fusion emerged as the most robust strategy, achieving success rates of 61.0% on LIBERO-Object and 74.8% on LIBERO-Spatial. By preserving the vision encoder, it maintains essential semantic features, in contrast to VGGT-Only, which caused a performance collapse (dropping to 7.2% and 15.2%). This confirms that geometric tokens cannot simply replace the rich texture and semantic cues of 2D representations. Interestingly, VGGT-Pointmap displayed high task sensitivity: it performed reasonably well on LIBERO-Spatial (41.4%) but failed on LIBERO-Object (9.0%).

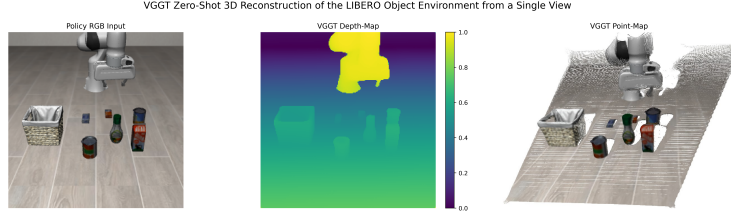


Figure 3: **VGGT Zero-Shot 3D Reconstruction from a Single View.** From a single RGB frame from the policy’s perspective (left), a frozen VGGT model produces a coherent metric depth map (center) and a dense pointmap (right, rendered from a side view). The reconstruction accurately captures the 3D structure of the scene, validating that the geometric tokens provided to Geo-Octo variants are high-quality and contain task-relevant information.

Despite these behavioral insights, no variant outperformed the Octo-Base baseline (70.8% and 82.0%). This indicates that under standard finetuning, the stability of the pretrained 2D features outweighs the added geometry, which appears to introduce interference rather than complementary guidance.

4.3 Analysis of Performance

To understand the reasons behind these results, we conducted a series of diagnostic experiments that we analyze in the remainder of this section.

Verifying the Quality of VGGT’s Geometric Cues. First, we aimed to confirm that the geometric information from VGGT is indeed accurate and useful for manipulation. As shown in Figure 3, VGGT can produce detailed and coherent 3D reconstructions from a single RGB frame in a zero-shot manner. The depth map correctly identifies the relative distances of objects, and the pointmap reconstructs their volumetric structure, capturing task-relevant geometry such as object shapes, heights, and support surfaces. This confirms that the input signal to our Geo-Octo models is rich with the exact spatial cues that we hypothesized would be beneficial. The failure, therefore, does not lie in the quality of the 3D information, but in its integration.

Diagnostic Analysis of Spatial Attention To investigate the mechanisms behind the integration failure of our methods we visualize the internal transformer attention weights. Specifically, we extract the cross-attention scores connecting the final action readout token to the spatial visual tokens at a critical decision-making timestep (grasping the ketchup bottle). This heatmap reveals where did the policy focus during the action prediction. Figure 4 presents this diagnostic analysis.

The baseline Octo model (Fig. 4, Left) exhibits a coherent attention pattern. The focus is distributed between the target trash can (the placement goal), and the area near the ketchup bottle, where the end-effector is about to perform a grasping action. While the attention on the goal is not a single pixel-perfect point, the general distribution covers the relevant semantic region. This confirms that the 2D-pretrained backbone, when operating in its native modality, maintains a robust understanding of the scene’s topology and task-relevant objects.

In contrast, the VGGT-Only model (Fig. 4, Right) displays a scattered and incoherent attention map. Despite successfully completing this specific episode (a statistical outlier given its 7.2% overall success rate), the underlying mechanism reveals a complete lack of spatial grounding. The attention is split between unrelated background features and the robot’s body. This evidence strongly suggests that the pretrained transformer possesses a powerful inductive bias towards 2D feature distributions and cannot reconcile them with the fundamentally different geometric statistics of the VGGT tokens. The policy effectively operates without true visual understanding, relying on spurious correlations that occasionally lead to successful actions.

The VGGT-Fusion variant presents a distinct behavior known as an *Attention Sink* (Xiao et al., 2024). As seen in the readout attention map (Fig. 4, Center Left), the model fixates almost entirely on a static, task-irrelevant patch on the floor. We observe this phenomenon in every attention map regardless of the timestep, the task, or the outcome of the task. We hypothesize that the distribution shift introduced by the fused 3D tokens creates high-entropy feature maps that overwhelm the pretrained attention heads. To minimize the impact of this noise on local object representations, the model learns to store

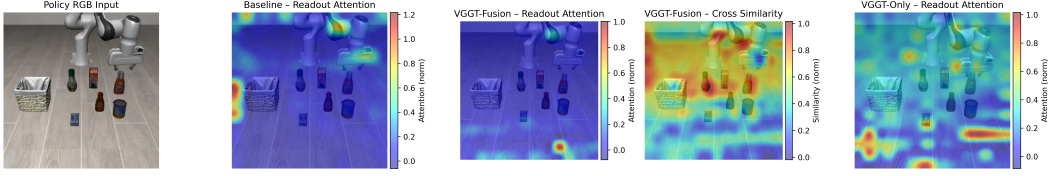


Figure 4: **Diagnostic Analysis of Spatial Attention and Feature Alignment.** Visualization of the policy’s internal focus during the task "pick the ketchup (top left object) and place it on the basket" (**Left**) The **Baseline** exhibits coherent attention on the gripper and target container. (**Right**) The **VGGT-Only** model shows scattered, hallucinated attention, indicative of spatial collapse. (**Center**) The **VGGT-Fusion** analysis reveals an *Attention Sink* phenomenon (Left), where the policy fixates on a background patch. The corresponding Cross-Similarity map (Right) shows that while feature alignment exists, it is diffuse and lacks the sharpness required for precise spatial attention.

the majority of its attention probability mass to a background sink token. This allows the model to ignore the corrupted local details while likely preserving a coarse, global context required for the task (explaining its moderate 61% success rate).

To confirm this, we supplement the analysis with a Cross-Similarity map (Fig. 4, Center Right), which measures the semantic alignment between the readout token and the feature patches. Unlike the collapsed attention map, the similarity map shows a diffuse but present alignment around the central objects. This confirms that the geometric information is quite similar with the information of the native encoder for the central objects, however it remains unknown whether there is any semantic understanding in this alignment. Future work could potentially mitigate this by introducing *register tokens* (Darcet et al., 2024), which would provide a dedicated container for this global context, allowing the spatial attention heads to refocus on local object details. The authors of this paper state that adding registers cleans the attention maps but also yields small performance gains. However, in our case, the benefit could be substantially bigger considering the difficulty of the data fusion.

Investigating Readout-Level Fusion with Pointmaps. The VGGT-Pointmap method integrated 3D information at the final readout stage, after the main transformer backbone. To analyze its effect, we examined how the action readout token’s focus shifts in response to the injected pointmap embedding. As shown in Figure 5, we computed the cosine similarity between the action readout vector and all image patch tokens, both with and without the 3D information.

The visualization shows that adding the 3D pointmap information does shift the readout’s focus towards the central workspace, confirming that the geometric signal has a tangible effect on the model’s final decision-making layer. However, this shift does not translate into a more precise focus on the target objects required by the language instruction. The saliency pattern remains broad and fails to prioritize the key objects. This suggests that even when bypassing the main vision-processing pipeline, the model’s pretrained representations are resistant to being effectively steered by external 3D cues. Finally, we underline that this approach is highly sensitive to hyperparameters (LoRA rank, learning rate), which we could not exhaustively tune. This sensitivity is further supported by the significant difference in performance between the two LIBERO splits.

5 Ablations

Beyond the strategies described in our main methodology, we explored several other variants to investigate the impact of our distinct components. Due to computational constraints, these experiments were limited, but are recorded to document the design space and provide insights for future work.

Spatial-Axis Concatenation. Our initial intuition for fusing intermediate tokens was to append them along the spatial axis, doubling the token count from 256 to 512 while keeping the feature width at 512. This approach, however, fundamentally alters the sequence structure that Octo’s transformer backbone expects. Because the model’s positional embeddings, readout mechanisms, and attention patterns are all pretrained on a 256-token vision stream, introducing a 512-token sequence during finetuning, resulted in a catastrophic regression. The model never recovered its pretrained competence,

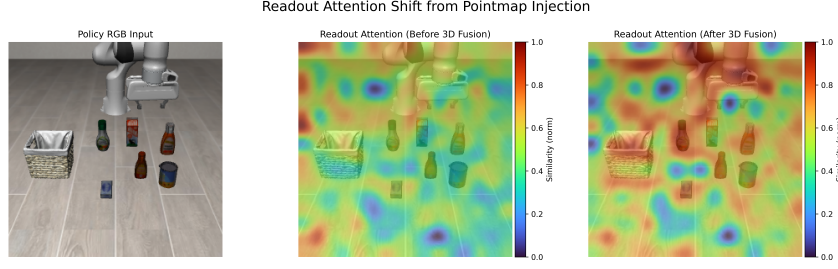


Figure 5: **Analysis of Readout Attention with Pointmap Fusion.** The heatmaps show the similarity between the action readout vector and the image patch tokens. **(Left)** Without 3D information, the readout has a baseline focus. **(Right)** After fusing the pointmap embedding, the readout’s focus shifts towards the central objects but fails to converge on the task-relevant objects. This indicates that while the 3D signal has an effect, it is not effectively bound to the language-conditioned goal.

and frequent loss spikes suggested that the attention blocks were effectively processing out-of-range indices. In none of our model checkpoints did this variant manage to surpass 10% accuracy in either LIBERO-Spatial or LIBERO-Object. We conclude that without re-pretraining the transformer on longer sequences, altering the spatial token count is an untenable strategy for lightweight finetuning.

Principal Component Analysis Compressor. We investigated replacing the autoencoder with Principal Component Analysis (PCA) as a non-learned dimensionality reduction technique. We first reshaped the VGGT features to (256, 2048), through a standard mean pooling, and subsequently fit PCA to find the top 512 principal components. Despite this linear projection preserved approximately 99.5% of the variance, the performance was inferior to the autoencoder baseline. In addition, this approach presented two critical issues: (i) fitting PCA on millions of high-dimensional vectors was more time-consuming compared to the autoencoder, and (ii) it could not perform the crucial weighted mean pooling across VGGT layers that our autoencoder handles. Consequently, the PCA route was both slower to prepare and weaker in performance, reinforcing our choice of a learned compressor.

Layer-Subset Aggregation. The authors of VGGT employ of a specific subset of layers for DPT prediction (specifically the 4-th, 11-th, 17-th, and 23-rd). Following their approach, we conducted an ablation using only this sparse subset of four informative layers instead of the full 24. While this configuration drastically reduced the computational and memory footprint of the feature extraction, the resulting policy consistently underperformed the one using the full 24-layer average. For LIBERO-Spatial, the best checkpoint for this method yields an accuracy of 52%. We did not test the impact of this variant on the LIBERO-Object split due to computational restrictions. To conclude, we hypothesize that collapsing to only four layers removes fine-grained geometric details that our autoencoder learns to exploit, and thus our final models retain information from all layers.

6 Conclusion

In this work we tested whether a 2D-pretrained VLA model, Octo, could benefit from geometric knowledge from a specialist vision model, VGGT, via three token-level finetuning schemes: fusion, replacement, and pointmap injection at the readout. Our evaluation on LIBERO highlights the resilience of the pretrained 2D representations, which consistently rejected the injected geometric signals. Through detailed attention analysis, we identified that this resistance manifests as attention sinks and spatial-semantic misalignment, suggesting that the modality gap cannot be bridged by architectural changes and simple finetuning. We believe meaningful gains would require integrating 3D signals earlier (e.g., during pretraining) so the model can learn a shared 2D/3D latent space.

Looking forward, we believe VGGT-Pointmap could benefit from a broader hyperparameter sweep that could stabilize the 3D injection. Finally, for VGGT-Fusion, the model seems to struggle to store global geometry alongside local semantics. Future work could address this by training with registers, which would provide dedicated capacity for 3D context without overwriting visual cues.

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